

Session 9 — Last night: first frames

Practical Astronomy Intensive

Name _____

Date _____

6 questions 20 minutes

ISO as gain not sensitivity, the 500 rule as a guide not a law, and trails as Earth turning under a fixed camera. Two of these are shot on the night — the histogram and the intervalometer run are the honest checks.

*Circle one letter for each multiple-choice question. Write short answers on the lines. Tasks marked **Practical** are done, not written — your assessor will watch.*

- 1 Your 20-second frame of the Milky Way is too dark. A friend says "just push the ISO to 12800, that makes the sensor more sensitive." Is he right?
 - a No — just leave the shutter open for two minutes instead, which collects genuinely more light
 - b Yes — higher ISO makes the sensor physically more sensitive, so it catches more light
 - c No — ISO amplifies the signal after capture, so it brightens the noise right along with the stars; the sensor collected the same photons either way
 - d No — instead open the aperture as wide as it goes, which is always the better move
- 2 The 500 rule says: max seconds before trailing = $500 / \text{focal length}$. On a 20mm lens that gives 25 seconds. How should you treat that number tonight?
 - a As applying equally everywhere in the frame, so if the centre is clean the corners are too
 - b As a rough film-era guide — on a modern high-resolution sensor stars trail at roughly half that, which is why the tighter NPF rule exists; shoot it and zoom in to check
 - c As exact — it is a rule, so 25 seconds is the precise threshold
 - d As a floor you can exceed by lowering the ISO, which keeps the stars from trailing
- 3 You want two hours of star trails. What do you actually shoot, and what will the frame show?
 - a Hundreds of short frames stacked, aimed anywhere you like — the arcs come out as concentric rings whatever direction you point
 - b One two-hour exposure in bulb mode — the trails need the shutter open the whole time to draw continuously
 - c Hundreds of short frames stacked — and the arcs record the stars themselves slowly moving across the sky
 - d Hundreds of 30-second frames back to back, blended later in lighten mode — one two-hour exposure just fogs on skyglow; and the arcs are Earth turning under a fixed camera, which is why they curve around the celestial pole

- 4 Frame and shoot one 20-second nightscape — sky over a real foreground — then bring up its histogram and tell the instructor what it says.

Practical — performed for your assessor.

- 5 You are shooting a nightscape at 20mm on a fixed tripod and the frame is a stop underexposed. Aperture is already at f/2.8 (the lens opens to f/1.8), shutter is at the trailing limit, ISO is 1600. Which move costs you least?
- a Raise the ISO a stop, since a more sensitive sensor gathers the missing photons cleanly
 - b Raise the ISO a stop — it adds noise but keeps the stars round; the shutter is capped by Earth's rotation and opening to f/1.8 would smear the corner stars into coma
 - c Open up to f/1.8 — the lens's widest setting is always its best setting
 - d Double the shutter to 40 seconds — more time is always more light

- 6 Set up a star-trail sequence: find Polaris by eye, frame it with a foreground, and start an intervalometer run of 30-second subs. Before you walk away, tell the instructor what shape the trails will take and why.

Practical — performed for your assessor.